

	conductivity of the semiconductor increases. Hence resistance decreases.
6(a)(i)	If the daughter is radiative, it will decay and the count rate measured will not be due to the decay of phosphorus-32 alone.
6(a)(ii)	Background radiation is less than 1 count per second, which is negligible compared to the few hundreds per second measured in this experiment. The percentage error is negligible.
6(b)	First half life = 15.25 days. Second half life = 14 days. Average half life = $(15.25+14)/2 = 14.6 \approx 15$ days
6(c)	Beta radiation does not cause immediate damage, but it can penetrate through human skin to ionise cells. This can lead to DNA mutation which can develop through time into cancer.
7(a)(i)1.	Displacement is defined as the linear distance moved by an object in a specified